

Time Series

2026/2027, Autumn Trimester

Course faculty	Ihor Miroshnychenko e-mail: imiroshnychenko@kse.org.ua
Department	Computer Science Department
Study program	Artificial Intelligence
ECTS credits	4 (120 hours)
Class hours	18 lecture hours and 44 practice hours (62 contact hours, 58 hours of self-study)
Course language	English/Ukrainian
Course format	Offline
Form of study	Full-time studies

Overview

This course comprehensively introduces time series analysis and forecasting using R or Python. Students will learn classical statistical methods (ETS, ARIMA, SARIMA, SARIMAX), machine learning approaches, and modern hybrid techniques for forecasting.

The course is built around in-class work. Every practice session is a working session: you build, evaluate and submit your solution within the class period, and you receive feedback on it before the next session. From Week 4 onward, most practice sessions take the form of an in-class forecasting challenge — you are assigned your own time series and must produce a forecast that beats a benchmark within the time available. The course culminates in a final project built on the same mechanic, using a shared high-frequency dataset.

Prerequisites

Dependent courses.

- Must have: Probability Theory and Statistics
- Nice to have: Courses in data analysis or machine learning

Hard skills.

- Must have: Proficiency in R or Python. Understanding of basic statistical concepts (mean, variance, correlation).
- Nice to have: Experience working with libraries for data analysis.

Soft skills. Problem-solving skills, analytical thinking, and the ability to work with data under time constraints.

Background and course rationale

Time series forecasting is a key data science skill used in finance, economics, energy, supply chain management, and many other domains. This course equips students with both theoretical understanding and practical skills to model and forecast time-dependent data.

This course differs from others in that:

- It uses R or Python, free, open source, and extremely powerful software.
- It includes dozens of real-world examples.
- It emphasizes graphical methods for exploring data, analyzing the reliability of fitted models, and presenting forecasting results.
- **It is practised under realistic constraints.** Forecasting in industry happens against a deadline and is judged against a benchmark. This course reproduces those conditions deliberately, and teaches that a more complex model is not automatically a better one.

This course addresses questions such as “How to analyze time series data?”, “What forecasting methods to use?” and “How to evaluate the accuracy of a forecast?”, providing students with practical skills in the R or Python environment.

Course aims

1. To familiarize students with the basic principles and methods of forecasting, and provide sufficient information for their effective use.
2. To teach students to apply various forecasting methods using R or Python tools.

3. To develop skills in analyzing time series data and evaluating forecast accuracy in the R or Python environment.
4. To prepare students to solve real-world forecasting problems in business and other fields using a practical, example-driven approach.
5. To inspire students to explore further more complex theoretical details and the latest developments in forecasting, including neural networks and foundation forecasting models.

Learning outcomes

By the end of the course, students will be able to:

1. Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting.
2. Apply various forecasting methods to time series data using R or Python libraries.
3. Analyze time series data using graphical methods for research, model validation, and presentation of results.
4. Specify a forecasting model manually from diagnostic evidence (ACF/PACF, differencing, seasonality) and justify that specification against what automatic model selection produces.
5. Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
6. Judge when additional model complexity is warranted and when it is not – including recognising cases where a simple benchmark cannot be beaten.
7. Apply modern techniques such as neural networks and foundation models to forecasting tasks.

Methods of Teaching and Assessment

The course includes:

- **Lectures.** Theoretical material and core methods. One two-hour lecture per week in Weeks 1–9. Slides are available on Moodle. **Attendance is strongly recommended;** the in-class format assumes you arrive already familiar with the week's reading (see *Self-study*).
- **Practice sessions.** All assessed coursework is produced during practice sessions. There are no separate defence sessions for weekly assignments.
- **In-class challenges.** From Week 4, most practice sessions are timed forecasting challenges on an individually assigned dataset.
- **Mid-term quiz.** A single proctored quiz taken electronically on Moodle.
- **Final project.** A two-session individual project on a shared dataset, followed by a written report and a public defence.

Course Faculty



Ihor Miroshnychenko, Ph.D. in Economics

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Career Highlights:

- Teacher, Faculty of Computer Science, KSE
- **Associate Professor**, Department of Technology Management, FIT, KNU
- **Lecturer**, MBA, International Institute of Business

Research Interests:

- Data Science
- Statistics
- Data Analysis
- Machine Learning

Teaching Philosophy:

I believe that data science is a powerful tool for understanding the world around us. My goal is to help students develop a deep appreciation for the role of data in decision-making and to equip them with the skills to analyze data effectively and communicate their findings clearly.

Personal Interests:

-  Guitar playing
-  Traveling
-  Photography
-  Gaming
-  Reading
-  Football

Interesting Facts:

-  Big fan of the Middle-earth universe.
-  Engaged in historical reconstruction
-  Metalhead

Course Structure

- **Duration:** 4 ECTS credits (120 hours), Autumn Trimester 2026/2027.
- **Weeks 1–9:** one lecture (2 hours) and one practice session (4 hours) per week.
- **Week 10:** two practice sessions (4 hours each), held **on different days, at least two days apart**.
- **Total:** 9 lectures (18 hours), 11 practice sessions (44 hours), 58 hours of self-study.

Grading Policy

General Grading Scheme

The course grade is the sum of all activities in the table below, capped at 100 points.

<i>Task</i>	<i>Count</i>	<i>Points per item</i>	<i>Total Points</i>
Weekly assignments	8	7	56
Mid-term quiz	1	12	12
Final project (challenge + report + defence)	1	32	32
Total			100
Participation (extra)	1	5	5

Successful completion of the final project is mandatory to pass the course. A student who does not submit the final project, or whose project receives a failing grade, is not eligible for a passing grade in the course, irrespective of performance on any other assessment.

Weekly in-class assignments

There are eight weekly assignments, each worth up to 7 points (56 points total). **All of them are produced and submitted during the practice session.** Work is submitted to Moodle before the end of the session; it is marked and commented on afterwards. There are no defence sessions and no separate homework submission.

Assignments run in two tracks.

Track A – Foundational labs (Weeks 1–3)

Labs 1–3 use **named, domain-rich datasets** drawn from fpp3 / tsibbledata (e.g. aus_retail, vic_elec, pedestrian, PBS). The domain is visible and the history has meaning, so the work is about reading the data, not about scoring it. These labs are not competitive: at this point in the course there is no forecasting method or accuracy metric to compete on.

Criteria	Points
Data preparation	1
EDA and visualisation	3

Implementation	2
Code quality and reproducibility	1
Total	7

Track B – In-class forecasting challenges (Weeks 4, 6, 7, 8, 9)

Labs 4–8 are timed challenges. **Each student is assigned their own time series**, drawn deterministically from a curated pool by a hash of the student identifier and the lab number. Series are taken from the M3 and M4 forecasting competition archives and are anonymised on issue: identifiers removed, values rescaled by a random constant, calendar shifted, history trimmed.

Because every student works on a different series, discussing methods with your peers cannot become copying a result. Such discussion is explicitly encouraged and is rewarded through participation points.

Criteria	Points
Forecast accuracy – beating the benchmark	3
Justification block	2
EDA and visualisation	1
Code quality and reproducibility	1
Total	7

How the benchmark works.

- Accuracy is measured by a **scale-free skill score**: MASE, or equivalently the ratio of your model's RMSE to the benchmark's RMSE on the hidden horizon. This makes results comparable across different series.
- The **benchmark** is the best of the simple methods – naive, seasonal naive, drift, and mean – selected on a validation window. It is not fixed in advance to seasonal naive, because seasonal naive is trivially weak on some series and unbeatable on others.
- The pool of series is **pre-screened**: a series is only issued if a reference AutoETS/AutoARIMA model achieves a skill score between 0.70 and 0.90 on it. Below that band the series is trivial; above it, effectively unbeatable. **Every series you may be assigned is one that is known to be beatable.** If you cannot beat the benchmark, that is a fact about your model, not about your luck.

Deliverables for each challenge lab: a submission CSV of point forecasts (and prediction intervals where the task requires them), your code, and the justification block.

The justification block consists of three short fixed questions, specific to the week's topic. From Week 6 onward the task is structured so that **you must specify the model manually first and justify that specification, and only then compare it with automatic model selection** and explain any divergence. Automatic selection cannot be your first step.

Feedback. The automated accuracy score is published **on the same day**. Rubric marks and written comments are returned **before the next practice session**, so there is always time to act on them.

Mid-term quiz

A single proctored quiz, held in the Week 5 practice slot and taken electronically on Moodle. It covers Chapters 1–3, 5 and 8 – the material completed before the quiz. Worth 12 points.

Final project

The final project is an **individual** assessment in which students act as data scientists for an energy utility company. All students work with the same standardised high-frequency (15-minute) time series dataset provided by the instructor. Students do not select their own dataset.

The project runs across the two Week 10 practice sessions, held on different days at least two days apart.

Session 1 – In-class prediction challenge. Within the class period, produce point forecasts and 95% prediction intervals for a hidden 48-hour test horizon. Submissions are scored on accuracy (RMSE, MAPE) and uncertainty quantification (Winkler score), against a seasonal naive benchmark.

Between sessions. Improve your model at home. Explore alternative approaches, tune, and prepare your findings.

Session 2 – Defence. Present your approach and results to the group. This session is a public exchange: you will see how others attacked the same dataset, which methods worked, and which did not.

Deliverables: the in-class submission CSV, the improved submission CSV, a written report (Jupyter Notebook or PDF), reproducible code (GitHub or archive), and presentation slides.

Criteria and point breakdown (total: 32 points)

Criteria	Points	Stage
Valid in-class submission – correct format, submitted on time	5	Session 1
Beating the benchmark in class	8	Session 1
Final skill score (class ranking)	5	Between sessions
EDA, methodology and diagnostics – residual analysis with ACF/PACF is required, not optional	8	Report
Code quality and reproducibility – must run without errors	2	Report
Defence and presentation	4	Session 2
Total	32	

On the final skill score. These 5 points are awarded on your absolute final accuracy on the shared dataset, ranked against the class – not on how much you improved relative to your own in-class

result. Two gates apply: an in-class submission must have been made, and your final result must be no worse than it. Scoring improvement rather than absolute accuracy would reward deliberately underperforming in class, which is not what this course is for.

Participation

Participation points reward **sharing what you have learned**, not merely being present. Attendance is already structurally required by the in-class format and is not paid for twice. Points are awarded for:

- explaining your approach to the group;
- helping a peer reason about their own assigned series;
- asking substantive questions and offering insights during lectures and defences;
- constructive peer interaction during classwork.

Bonus points are added on top of your final grade, subject to the 100-point cap. Consistent disengagement results in 0 bonus points.

Attendance, missed sessions and interruptions

Because all coursework is produced in class, absence has a direct cost. The following provisions exist so that circumstances outside your control do not decide your grade.

Situation	Provision
Missed weekly assignment	An asynchronous replacement is issued: the same dataset, deadline 24 hours later. Capped at 5 of 7 points , no accuracy bonus. Available at most twice per trimester without documentation; further use requires documented grounds.
Air raid alert during a practice session	If the interruption exceeds 20 minutes, the session converts to asynchronous mode for the entire group, with no penalty and no cap.
Missed Week 10 challenge (Session 1)	A supervised make-up is arranged within one week on documented grounds: the same dataset, the same 48-hour horizon, the same 90 minutes of working time, in the presence of the instructor. Base points are awarded in full; the class-ranking bonus is not awarded, since the class distribution is by then known.
Air raid alert during Week 10, Session 1	If the interruption exceeds 20 minutes, the challenge is rescheduled in full for the entire group to a reserve date. It is not converted to asynchronous mode: this session gates passing the course.

Because coursework is no longer submitted as homework, the previous late-submission scheme (three days at -20% per day) no longer applies and is replaced by the provisions above.

Self-study (58 hours)

Self-study is a required component of the course, not optional reading. The in-class format assumes you arrive at the practice session already familiar with the week's method; four hours is enough to apply a method you have read about, and not enough to learn one from scratch.

Activity	Hours
Required pre-reading of the assigned chapter before each lecture (Weeks 1–9, ~3 h/week)	27
Final project – home improvement phase between Week 10 sessions	13
Final project – report and presentation slides	10
Mid-term quiz preparation	8
Total	58

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We do not tolerate plagiarism and cheating. Very competitive essence of the program calls for absolutely equal treatment of all students. Even a single case of violation of academic integrity is a serious misdemeanor which may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions and other merit-based awards.

The minimum penalty for a student caught cheating or plagiarizing during any practical task, a presentation or group and final projects will be a zero for this assignment with the consecutive decrease of the final course grade by at least one letter-grade. Repetitive violation of academic integrity may lead to the dismissal of the student from KSE. By enrolling in this course, students affirm that they are familiar with the KSE Code of Academic Integrity.

AI Tools usage policy

Scope. You are encouraged to explore and use generative AI as a learning tool, similar to how you would use a library, a search engine, or a discussion with a peer. You are ultimately responsible for the content and originality of all submitted work.

AI for coding assistance – permitted. You may use AI tools for coding: generating boilerplate, debugging, understanding concepts, or exploring alternative implementations. This includes work done during in-class challenges.

Citation.

- *In-class work:* prompt-level citation is **not** required. Citing prompts under time pressure is not enforceable, and requiring it would only produce a rule that everyone breaks. The **justification block** replaces it: for every challenge submission you must explain, in your own words, what

you did and why. It is worth 2 of the 7 points, it is marked by hand, and you may be asked follow-up questions about it in the next session.

- *Final project report*: citation **is** required. State the tool used and the prompts that generated any incorporated code.

AI for narrative and analysis — not permitted. Unless explicitly permitted for a specific assignment, you may not use generative AI to write the narrative portions of your work: written analyses, reports, explanations of your results, the justification blocks, or any text meant to represent your own understanding and interpretation.

Penalties. Any violation of this policy is treated as a breach of academic integrity, equivalent to plagiarism. Consequences may include a grade of zero on the assignment, failure of the course, and referral to the Academic Integrity Committee.

By submitting work in this course, you affirm that you have complied with this policy. If you have any questions about what constitutes appropriate use of AI, please ask the instructor before submitting your work.

Course Plan

Learning materials

Books:

1. Hyndman, R.J., & Athanasopoulos, G. (2021). Forecasting: principles and practice, 3rd edition, OTexts: Melbourne, Australia. [OTexts.com/fpp3](https://otexts.com/fpp3).
2. Hyndman, R.J., Athanasopoulos, G., Garza, A., Challu, C., Mergenthaler, M., & Olivares, K.G. (2024). Forecasting: Principles and Practice, the Pythonic Way. OTexts: Melbourne, Australia. Available at: [OTexts.com/fpppy](https://otexts.com/fpppy)

Schedule

Week	Topics	Learning Materials	Learning Outcomes
1	Introduction to forecasting + Time series graphics.		
	• Course overview; • basic concepts of forecasting; • the forecaster's mindset; • time plots, seasonal and subseries plots, lag plots; • autocorrelation and white noise	Forecasting: principles and practice book (Ch. 1–2)	• Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting. • Analyze time series data using graphical methods for research, model validation, and presentation of results.
2	Time series decomposition.		

	• Transformations and adjustments; time series components; • moving averages; • classical decomposition; • STL decomposition.	Forecasting: principles and practice book (Chapter 3)	• Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting. • Analyze time series data using graphical methods for research, model validation, and presentation of results.
3	The forecaster's toolbox.		
	• Forecasting workflow; • simple forecasting methods; • fitted values and residuals; • residual diagnostics; • distributional forecasts and prediction intervals; • forecasting with transformations and with decomposition; • evaluating point forecast accuracy; • time series cross-validation.	Forecasting: principles and practice book (Chapter 5)	• Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting. • Analyze time series data using graphical methods for research, model validation, and presentation of results.
4	Exponential smoothing.		
	• Simple exponential smoothing; • methods with trend; • methods with seasonality; • a taxonomy of exponential	Forecasting: principles and practice book (Chapter 8)	• Apply various forecasting methods to time series data using R or Python libraries.

	smoothing methods; • ETS models; • estimation and model selection; • forecasting with ETS.		• Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
5	Time series regression models (bonus topic) Mid-term Quiz		
	• Linear models for time series; • trend and seasonal dummies; • selecting predictors; • forecasting with regression	Forecasting: principles and practice book (Chapter 7)	• Apply various forecasting methods to time series data using R or Python libraries. • Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
6	ARIMA models, part 1.		
	• Stationarity and differencing; • autoregressive models; • moving average models; • non-seasonal ARIMA; • manual order identification from ACF/PACF; • estimation.	Forecasting: principles and practice book (Chapter 9)	• Apply various forecasting methods to time series data using R or Python libraries. • Specify a forecasting model manually from diagnostic evidence (ACF/PACF, differencing, seasonality) and justify that

			specification against what automatic model selection produces. • Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
7	ARIMA models, part 2.		
	• Seasonal ARIMA; • automatic order selection; • ARIMA vs ETS; • SARIMAX / dynamic regression models	Forecasting: principles and practice book (Chapter 9-10)	• Apply various forecasting methods to time series data using R or Python libraries. • Specify a forecasting model manually from diagnostic evidence (ACF/PACF, differencing, seasonality) and justify that specification against what automatic model selection produces. • Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
8	Advanced forecasting methods.		

	• Complex seasonality; • Prophet; • vector autoregressions; • bootstrapping and bagging.	Forecasting: principles and practice book (Chapter 12)	• Judge when additional model complexity is warranted and when it is not – including recognising cases where a simple benchmark cannot be beaten. • Apply modern techniques such as neural networks and foundation models to forecasting tasks.
9	Neural network forecasting.		
	• Neural forecasting; multilayer perceptron; • modern architectures; • optimization objectives; • exogenous variables; • hyperparameter optimization; • transfer learning; • overview of foundation models.	Forecasting: principles and practice book (Chapter 14-15)	• Judge when additional model complexity is warranted and when it is not – including recognising cases where a simple benchmark cannot be beaten. • Apply modern techniques such as neural networks and foundation models to forecasting tasks.
10	Finale Project		
	• Session 1: final in-class prediction challenge (shared dataset, 48-hour hidden horizon). • Session 2 (different day, ≥ 2 days later): defence and		• Apply various forecasting methods to time series data using R or Python libraries. • Evaluate the accuracy of forecasts against a benchmark, quantify forecast

	exchange of approaches.		uncertainty, and select the most appropriate model for a specific task. • Judge when additional model complexity is warranted and when it is not – including recognising cases where a simple benchmark cannot be beaten. • Apply modern techniques such as neural networks and foundation models to forecasting tasks.
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